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Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

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D7.2 Plan for Dissemination and Exploitation

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1 Introduction

This document serves as a first effort to provide a comprehensive dissemination and exploitation strategy for the SymAware project. This is a "living" document which will be updated at M12 and M36 to include the completed, ongoing, and planned results of the dissemination and exploitation activities, the evaluation of the dissemination and exploitation strategies and the actions taken to improve it. This allows us to adapt our activities based on the project progress, intermediate results and opportunities that arise on the outside.

The deliverable consists of two main parts: plan for dissemination activities and plan for exploitation activities. The contents of the deliverable have been divided into 7 sections:

1. Introduction: A brief description of the deliverable, as well as the purpose of the document. logical characterization of awareness using symbolic methods,

2. Dissemination strategy and approach: An overview of the strategy to be followed to disseminate the results and actions of the SymAware project; A description on the initial dissemination plan, describing the key messages and scientific areas, the target audience of the SymAware dissemination strategy, the channels, and the expected results of the plan.

3. Completed and on-going dissemination activities: A detailed list of the completed and on-going dissemination activities and developments following the dissemination strategies.

4. Future dissemination activities: A plan for new activities to be conducted for future developments during the life of the project.

5. Completed and on-going exploitation activities.

6. Future exploitation activities.

2 Dissemination strategy and approach

Dissemination activities are key in SymAware and they were managed by asking the collaboration of all partners. This demonstrates a strong commitment to raise awareness of the project, to different target audiences. Efficient dissemination is fundamental in any research process, as the success of projects outcomes is largely connected to dissemination activities. Careful and early planning of dissemination activities therefore is very important.

A dissemination strategy was created to be carried out along with the main scientific outputs of the project. The dissemination plan breaks down 36 months of the project into 3 reports that categorize the strategic activities completed. For each dissemination report, events are classified into appearances, publications, and promotional material within the stated 6 months. Information regarding the distributor, date, event, activity, the targeted group and attendee are provided in chronological order. Additional information with the link and a short description of the event is included in the chart for quick reference. In addition, a dissemination plan for the next 12 months is provided at the end of the document.

The dissemination of SymAware is based on achievement of work packages W1 to W8 from months 1 to 36 of the project. These work packages are:

WP1: Situational Awareness
WP2: Spatiotemporal goal awareness
WP3: Risk Awareness
WP4: Knowledge awareness
WP5: Use cases
WP6: Requirements for trustworthy and ethical human-agent interaction
WP7: Exploitation, Communication and Dissemination
WP8: Project Management

Dissemination activities will be achieved and managed as part of WP7. Activities in this work package demonstrate the commitment of the consortium group in raising awareness of the project to the target markets. The main objectives of this WP are:

- Reporting and archiving results
- Constructing and updating the project website
- Promoting scientific dissemination
- Supporting intellectual property issues and commercialization

The deliverables for exploitation and dissemination under WP7 includes:

- D7.1 (M6) “Plan for Dissemination and Exploitation (first version)”
- D7.2 (M12) “Plan for Dissemination and Exploitation (second version)”
- D7.3 (M36) “Plan for Dissemination and Exploitation (final version)”
- D7.4 (M6) “Data Management Plan (first version)”.
- D7.5 (M36) “Data Management Plan (final version)”.
- D7.6 (M3) “Public website”, Design and first implementation with core (static) content.

- D7.7 (M12) “Software tools on public Git repository (first release)”.
- D7.8 (M24) “Software tools on public Git repository (second release)”.
- D7.9 (M34) “Software tools on public Git repository (third release)”.

Making the project and its deliverables attractive to stakeholders is a key part of SymAware and it is important to create clear activities that help to spread community awareness of the scientific and technical aspects of the project. The dissemination strategy for SymAware will include participation in events, stakeholder liaison, online activity, media and scientific dissemination. Different communication channels will be used in order to address all target groups. Efficient dissemination is a fundamental activity in any research process, as the success of dissemination activities contributes to project outcomes by all interested target groups.

Dissemination activities will be managed as part of WP7, where all project partners will be asked to contribute. The project consortium demonstrated commitment in this sense to raise awareness of the project. The following important actions were taken to maximize the impact of dissemination:

- identify the relevant target groups
- share the SymAware project idea in the most suitable way according to the needs of target groups
- identify any barriers in order to be able to make sure dissemination goes beyond these
- deliver dissemination materials and activities

Dissemination strategy

During the project, efforts will be made to disseminate project goals and results to a variety of groups, this includes the writing of scientific publications to communicate the results to the scientific community. Conferences and seminars will be planned and attended to target researchers, and other key audiences in the scientific communities. Several measures will be taken in order to disseminate results to target groups. Scientific publications will be written and sent to journals in order to communicate the results to the scientific community. Consortium members will be attending events and seminars to target stakeholder groups in the scientific community.

Papers, presentations and other dissemination activities

Presentations and papers were prepared by consortium partners that attended specific conferences and events. In particular:

- Workshops at…
- Conference papers presented at…
- Scientific papers…
- Other activities (including keynote speeches, exhibitions and conferences)

Further reports and scientific papers are to be published throughout each stage of the project, up until the project end, at M36.

SymAware Dissemination strategy

<u>Information to disseminate</u>

- SymAware goals and potential exploitable results

<u>Dissemination</u>

1. Scientific publications and software tools
   - Publications
   - Technical papers
   - Presentations
   - Reports
   - White papers
   - Development of software tools
2. Social media visibility and project website
- Online activity
  - Blog posts about ongoing activities and results
  - Mention of activities on social media channels
  - Email newsletters
  - Project website development
- Media
  - Press releases about the project, updates or milestones
  - Interviews
3. SymAware community building events
- Participation in events (exhibitions, meetings)
  - Explanation of the project
  - Presentations, keynote speeches, etc
  - Distribution of leaflets
- Liaison with stakeholders
  - Visits
  - Email updates
4. Portfolio activities
5. Advisory board

3 Completed and on-going dissemination activities

3.1 Scientific publications and software tools

The consortium contributed to a variety of scientific papers, publications and articles published during the period of the SymAware project. This included papers aimed at various different targets, such as the control community and robotics community.

See all of the Scientific Papers, Publications and articles published in Appendix A.1.

3.2 Social media visibility and project website

Promotional material distributed to support the dissemination of the SymAware project by consortium members included blogs, videos, websites and social media pages to mark various project milestones.

Project logo

An important aspect of establishing the project's identity was designing a project logo, which would be essential for use in dissemination material. A number of logos were designed and a project logo was agreed on by the consortium as follows.

Project website

The project website is a dissemination tool where key information about the project can be shared, for example details on members of the consortium, project objectives and dissemination activities such as publication of research papers. This provides key information for partners writing about or disseminating information about the project, a well as providing a source of information for external parties interested in finding out more about the project. The address for the website is https://www.symaware.eu/

The website has been updated regularly as the project progresses.

Social media tools

SymAware social media strategy aimed at providing stakeholders with information and new developments in the SymAware project. The aim of the posts is also to reach a wider audience. The social media platforms the project will use are LinkedIn, Twitter, and YouTube. SymAware has its own channels for each.

LinkedIn

SymAware has a LinkedIn account. All consortium members follow the SymAware LinkedIn page at: https://www.linkedin.com/company/symaware/ and share updates on their own LinkedIn accounts.

Twitter

All consortium members follow the SymAware Twitter page at: https://twitter.com/SymAware and repost messages on their own Twitter accounts. Sharing messages on the Twitter channel will involved sharing videos from the YouTube channel.

YouTube

SymAware has a YouTube channel at: https://www.youtube.com/@SymAwareProject/featured where relevant videos will been shared, including all project milestone videos, and project interviews and introductory videos.

Leaflet

SymAware produced a leaflet to serve dissemination purposes at events that the consortium participated in. The leaflet included details on the project and contact information.

Newsletter

SymAware distributes newsletters regularly. Registrants will be kept up to date of the project progress and the upcoming (online) workshops.

3.3 SymAware community building events

The SymAware consortium led and or participated in a range of events throughout the project period which helped to disseminate the goals and outcomes of the SymAware project. This included annual events, talks, workshops and demonstrations in Europe, the US and Asia, between 2022 and 2025. Annual events with a large participation from the scientific community attended by consortium members included:

At some of these events, SymAware was focused on particularly, for example in a presentation of project highlights and in related talks and presentations.

Events consortium members attended during M1-M6 included:

- SymAware Kickoff Meeting
- ACM/IEEE International Conference on Human-Robot Interaction (HRI)
- CPS Workshop
- 61st IEEE Conference on Decision and Control

Events consortium members attended during M7-M12 included:

- SymAware Consortium Meeting
- Dr Sadegh Soudjani attended the QEST conference and discussed the SymAware project activities with the formal methods community
- Dr Sadegh Soudjani organised an invited session at the FORMATS conference on scalable reachability analysis of dynamical systems

Find more information about events attended in Appendix A.2.

3.4 Portfolio activities

Portfolio activities consortium members attended during M1-M6 included:

- Portfolio kick-off meeting on 7th and 8th December 2022 at Saint-Josse-ten-Noode Brussels, Belgium
- Will Hendrix and Hiltje Nawijn from TUE had an online meeting with Renan Picoreti from EMERGE project to discuss issues related to communications activities in the portfolio.
- Sadegh Soudjani from MPI-SWS and Hiltje Nawijn from TUE had an online meeting with Isabelle Gordge, project manager of the EIC Pathfinder Astound project on possible collaboration to navigate the path from research project to business development, company building, and successful product commercialization.

Portfolio activities consortium members attended during M7-M12 included:

- The members of the SymAware project attended the yearly consortium meeting on 20–21 June 2023 at Eindhoven University of Technology in Eindhoven, the Netherlands
- Sadegh Soudjani from MPI-SWS attended the steering committee meeting with portfolio members on 12 April 2023 which was held online.
- Will Hendrix and Hiltje Nawijn from TUE participated in the kick-off meeting of the Awareness Inside working group on Communications which was held online on 28 June 2023
- Ana Tanevska from UU participated in the kick-off meeting of the Awareness Inside working group on Ethics, which was also held online on June 28th 2023.

- Ana Tanevska from UU and Arabinda Ghosh from MPI-SWS participated and presented the ethical issues in the SymAware project in the EIC Portfolio Ethics Meeting which was held on 20 September 2023 at Brussels, Belgium.
- Sadegh Soudjani from MPI-SWS and Lefteris Vlahakis from KTH will participate in a portfolio workshop titled “Awareness Inside: Open Meeting of the EIC Pathfinder Challenge” on 1 October 2023 in ECAI 2023 at Krakow, Poland.

3.5 Advisory board

Advisory board consortium members attended during M1-M6 included:

- Meetings will be arranged with the advisory board members to get the feedback on the progress of the project.
- MPI-SWS had a meeting with the independent ethical adviser (EA) on 16th March 2023
- UU had a meeting with the independent ethical adviser (EA) on 16th March 2023.

Advisory board consortium members attended during M7-M12 included:

- No activities during M7-M12.

4 Future dissemination activities

Partner	Exhibitions, Demonstrations, Forums, Seminars etc.	Papers, publications and articles	Promotional material	Media engagement	Other
Report M13-M36					
MPI-SWS	Dr Sadegh Soudjanin is organizing four invited session at the CDC 2023 conference on resilience, safety, and security of CPS in December 2023				
	Dr Sadegh Soudjani is one of the program committee members and the publicity chair of the ICCPS 2024 for Europe				
	Dr Sadegh Soudjani is one of the program committee members in the ADHS 2024 conference				
	Dr Sadegh Soudjani is in PhD defense committee members: of Birgit van Huijgevoort at TU Eindhoven in November, Ali Salamati at Ludwig Maximilian University Munich in October, Mahathi Anand at Ludwig Maximilian University Munich in September, Martin Sidaway in August at Newcastle University, and Anas Makdesi from Supelec in December.				
	Dr Sadegh Soudjani is giving a presentation at a workshop along ECAI in Krakow as part of the portfolio activities				
	Dr Sadegh Soudjani is giving a presentation at Karlsruhe Institute of Technology on data-driven methods				

	for verification and synthesis of cyber-physical systems				
TU/e	Dr. Sofie Haesaert will be presenting at the Conference on Decision and Control 2023 on data driven control				
KTH	Prof. Dr. Dimos Dimarogonas is organizing an invited session at the CDC 2023 conference on Formal Methods for Time-Critical Decision Making and Control				
	Prof. Dr. Dimos Dimarogonas will be presenting in the workshop on Networked Dynamical Systems: Assured Learning, Communication and Control in the University of Maryland				
UU					
NLR					
SISW					

5 Exploitation strategy and approach

5.1 Introduction

To maximize the chance of future uptake of project results, the consortium will empower key actors by building a SymAware community of followers, consisting of both academics specializing in related fields and especially young companies (start-up and scale-up phase) developing business in the field of autonomous systems. We will actively seek contact and stay in close communication with these stakeholders to (1) enable them to follow our developments, (2) provide them with releases of our developed software tools, (3) organize workshops with tutorials and examples during hands-on sessions using our developed software tools and (4) getting their feedback regarding the technology potential as they see it from their perspectives.

For building this community of followers we will disseminate the project details (goals, partners, structure) to a wide as possible target group of potentially interested parties (academics and industry) using many different communication channels. We will invite interested people to actively register as "SymAware follower".
This will result in a list of contact emails we will use for providing the community with:

- Updates on publications/presentation of SymAware researchers. In this way followers know at what Conference they could meet with SymAware researchers.
- Invitations to SymAware workshops. Workshops to work with SymAware developed software tools are scheduled in M12, M24 and M34.
- Newsletter to keep track of progress.

5.2 Building network of SymAware Followers

The target group of potential SymAware followers are academic groups in Computer Science and Control and industries with R&D staff profiles closely matching those of the SymAware researchers.
We will send out invitations to register as a SymAware follower using the following channels:

- All academic project partners using their own professional network.
- All industry project partners using their relevant customer contacts
- All portfolio partners partners which we will also ask to be ambassador of our project
- The invitation to register will be placed on our public website
- The invitation to register will be placed on the Portfolio website
- The invitation to register will be placed on our LinkedIn and Twitter channels
- The invitation to register will be placed on our project flyer that will distributed
- The invitation to register will be placed on all other dissemination and communications, where possible

5.3 Software releases and workshops

Key activity for increasing the changes for uptake of project results will be through actively guiding people in using the software tools that are based on the research outcomes of the project.

Three releases of the SymAware framework and tools integration have been scheduled in M12, M24 and M34. Soon after these release dates, workshops will be organized where SymAware followers can use the software framework under guidance of the SymAware researchers.

5.4 Exploitation plans from SymAware partners

Exploitation potential of project results for SymAware industrial partners will be identified and were appropriate further research directions could be taken such to further explore these exploitation potentials.

The same holds for researchers at academic partners that see potential to further develop project results into novel products or services through start-up companies.

6 Completed and on-going exploitation activities

6.1 Building network of SymAware Followers

The following activities have been completed:

- An online registration tool to collect follower registrations has been set up. The registration link is included in all social media communications and also has a prominent place on the SymAware website. So far, 10 followers have registered, of which 5 from industry.
- An online newsletter template to be able to send regular updates to the SymAware followers.
- A project flyer, which has been posted on the social media channels and handed out out during conferences/events.

The following activities are ongoing:

- Development of a promotional web page to give an outlook on project results and the potential use for industry.
- Social media campaign to attract more project followers
- Personal network campaign to attract more project followers

6.2 Software releases and workshops

The following activities have been completed:

- First release of the software tools on a public Github repository

The following activities are ongoing:

- Exploring possible workshop setup (form, content, location, connection to other events and so on)

6.3 Exploitation plans from SymAware partners

- No completed and ongoing plans

7 Future exploitation activities

7.1 Building network of SymAware Followers

- Preparation of followers newsletter
- Release of promotional web page for industry

7.2 Software releases and workshops

- Prepare for Software release 2 – adjusted approach
  The budget for WP1 didn't allow having a dedicated Research Software Engineer (RSE) for the full duration of the project. MPI-SWS is in the process of hiring an RSE that will be the main contact point for software integration during M13-M36. Once this person is in post, MPI-SWS will ask the partners to specify their contact persons and have a working group on "software integration" with regular meetings. MPI-SWS has intentionally held back this hiring to allow scientific approaches being developed in the first year and produce something to be integrated within the designed architecture. Although we have not produced in the first year a tool usable by external parties, the designed architecture and its initial implementation is the first milestone that achieves the objectives of creating an integrated framework for modeling, specifying logical requirements, and encoding awareness in decision making and control. This will specify a road map for future implementations in the project by the internal parties.

- Preparation of first followers workshop
  This task led by TU/e will be completed once an initial version of the integrated tool is ready to be used by external parties. The expected time line for this workshop is M19-M27 (second half of the second year or right after the second release of the tool). The current focus before the workshop will be to expand the community of followers. MPI-SWS is in contact with a previous EU project (HUBCAP) on software development for small and medium-sized enterprises to utilize their available community of followers and attract them to the SymAware project.

7.3 Exploitation plans from SymAware partners

Siemens

Verification and validation (V&V) of automated driving (AD) vehicles is an important topic for car OEMs to bring these vehicles into the market. One of the important safety standard for V&V of AD vehicles is ISO/PAS 21448 safety of intended functionality (SOTIF) standard. SOTIF categorizes all possible scenarios to 4 areas (1-known-safe,2-known-unsafe,3-unknown-unsafe, 4-unknown-safe). Main objective of SOTIF standard is on unknown-unsafe scenarios.

Siemens has developed a patented methodology called critical scenario creation (CSC) which could generate SOTIF's unknown-unsafe scenario through optimization algorithm. This methodology helps Siemens's customers to test their AD vehicles/functionalities in simulation environment for SOTIF scenarios and get safety approval.
Strengthening the CSC methodology to provide more challenging scenarios for advanced AD vehicles which are using situational awareness is the main goal of Siemens. CSC as an adversarial approach can generate more challenging scenarios if it knows how the future AD control algorithm based on awareness works and utilizes this knowledge in generation of the more critical scenarios.
Generating more challenging unknown-unsafe scenarios in simulation environment helps car OEMs to reduces significantly V&V process and therefore time to market.

NLR
SymAware supports NLR's programme "Unmanned and autonomous". Drones are a key enabler for innovative solutions to tackle rapidly changing societal, economic and security challenges. For the Netherlands to play a role in this, the government and Dutch industry, including start-ups and SMEs, must be able to create innovative drone solutions, platforms and systems in a safe, sustainable and cost-effective way, and to integrate these within the airspace. NLR offers support in the development, implementation and societal acceptance of unmanned and autonomous aviation, taking into account safety and minimal nuisance. NLR helps with the realisation of safe, sustainable and affordable drone systems and the harnessing of commercial opportunities, and supports the rise of Dutch manufacturers in the area of drones and Urban Air Mobility (UAM).

Appendix

A.1 Scientific publications and software tools

Partner	Title	Author	Year and Magazine	Target	Status	Joint publication
Report M01-M06						
MPI-SWS	Symbolic Knowledge Awareness for Autonomous Driving	Andrea Calvagna, Arabinda Ghosh, and Sadegh Soudjnai	2023	CDC 2023	Under preparation/ submission	No
	SySCoRe: Synthesis via Stochastic Coupling Relations	Birgit van Huijgevoort, Oliver Schön, Sadegh Soudjani, Sofie Haesaert	2023	HSCC 2023	Accepted	Yes
	Correct-by-Design Control of Parametric Stochastic Systems	Oliver Schön, Birgit van Huijgevoort, Sofie Haesaert, and Sadegh Soudjani	2022	CDC 2022	Accepted	Yes
	Data-Driven Abstraction-based Control Synthesis	Milad Kazemi, Rupak Majumdar, Mahmoud Salamati, Sadegh Soudjani, Ben Wooding	2023	Nonlinear Analysis Hybrid Systems	Under submission	Yes
	A Toolchain for Accelerated Symbolic Control (Software tool)	Rupak Majumdar, Kaushik Mallik, Mateusz	2023	HSCC	Accepted as poster	Yes

		Rychlicki, Anne-Kathrin Schmuck, Sadegh Soudjani				
	A Toolchain for Accelerated Symbolic Control (Software tool paper)	Rupak Majumdar, Kaushik Mallik, Mateusz Rychlicki, Anne-Kathrin Schmuck, Sadegh Soudjani	2023	CAV	Under submission	Yes
TU/e	SySCoRe: Synthesis via Stochastic Coupling Relations	Birgit van Huijgevoort, Oliver Schön, Sadegh Soudjani, Sofie Haesaert	2023	HSCC 2023	Accepted	Yes
	Correct-by-Design Control of Parametric Stochastic Systems	Oliver Schön, Birgit van Huijgevoort, Sofie Haesaert, and Sadegh Soudjani	2022	CDC 2022	Accepted	Yes
KTH	Distributed control synthesis of signal temporal logic tasks for multi-agent systems	S. Liu, A. Saoud, P. Jagtap, D. V. Dimarogonas, and M. Zamani.	2022	CDC2022	Accepted	No
	High-order control barrier functions for a prototype multi-spacecraft inspection of the ISS	Gregorio Marchesini, Pedro Roque, Dimos Dimarogonas	2023	CDC2023	Under preparation/ submission	No
UU	A Case Study in Designing Trustworthy Interactions and its Implications for Socially Assistive Robotics	Mengyu Zhong, Marc Fraile, Ginevra Castellano and Katie Winkle	2023	Frontiers	Accepted	No

NLR						
SISW						
Report M07-M12						
MPI-SWS	Data-Driven Correct-by-Design Control of Parametric Stochastic Systems	Oliver Schön, Birgit Van Huijgevoort, Sofie Haesaert, Sadegh Soudjani	2023	Proceedings of the 26th ACM International Conference on Hybrid Systems: Computation and Control	Published	Yes
	Robust Simulation Functions with Disturbance Refinement	Ben Wooding, Abolfazl Lavaei, Vahid Vahidinasab, Sadegh Soudjani	2023	2023 European Control Conference (ECC)	Published	No
	A Flexible Toolchain for Symbolic Rabin Games under Fair and Stochastic Uncertainties	Rupak Majumdar, Kaushik Mallik, Mateusz Rychlicki, Anne-Kathrin Schmuck, Sadegh Soudjani	2023	International Conference on Computer Aided Verification	Published	No
	Verifying the Unknown: Correct-by-Design Control Synthesis for Networks of Stochastic Uncertain Systems	Oliver Schön, Birgit van Huijgevoort, Sofie Haesaert, Sadegh Soudjani	2023	arXiv preprint arXiv:2309.01276	Under preparation/ submission	Yes
	Neural Abstraction-Based Controller Synthesis and Deployment	Rupak Majumdar, Mahmoud Salamati, Sadegh Soudjani	2023	ACM Transactions on Embedded Computing Systems	Published	No
	Safety Barrier Certificates	Omid	2023	arXiv preprint	Under	No

	for Stochastic Control Systems with Wireless Communication Networks	Akbarzadeh, Sadegh Soudjani, Abolfazl Lavaei		arXiv:2309.05570	preparation/ submission	
	Verification of Quantum Systems Using Barrier Certificates	Marco Lewis, Paolo Zuliani, Sadegh Soudjani	2023	International Conference on Quantitative Evaluation of Systems	Published	No
	Formal Control of New England 39-Bus Test System: An Assume-Guarantee Approach	Ben Wooding, Abolfazl Lavaei, Sadegh Soudjani	2023	arXiv preprint arXiv:2307.03467	Under preparation/ submission	No
	Using Knowledge Awareness to improve Safety of Autonomous Driving	Andrea Calvagna, Arabinda Ghosh, and Sadegh Soudjnai	2023	IEEE SMC 2023	Accepted	No
TU/e	Stochastic model predictive control with dynamic chance constraints	M. H. W. Engelaar, S. Haesaert, and M. Lazar	2023	CDC 2023	Accepted	No
	Modularized control synthesis for complex signal temporal logic specifications	Z. Zhang and S. Haesaert	2023	CDC 2023	Accepted	No
	Direct data-driven signal temporal logic control of linear systems	B. C. van Huijgevoort, C. Verhoek, R. Tóth, and S. Haesaert	2023	CDC 2023	Accepted	No
	Cautious planning with incremental symbolic perception: Designing verified reactive driving maneuvers	Kamale, Disha, Sofie Haesaert, and Cristian-Ioan Vasile	2023	ICRA 2023	Accepted	No
	Automated Formation Control Synthesis from	Shuhao Qi, Zengjie Zhang,	2023	CDC 2023	Accepted	No

	Temporal Logic Specifications	Sofie Haesaert, Zhiyong Sun				
	Risk-Aware Reward Shaping of Reinforcement Learning Agents for Autonomous Driving	Lin-Chi Wu, Zengjie Zhang, Zhiyong Sun, Zhiqiang Ma, Sofie Haesaert	2023	IECON 2023	Accepted	No
	Identifying Reaction-Aware Driving Styles of Stochastic Model Predictive Controlled Vehicles by Inverse Reinforcement Learning	Ni Dang, Tao Shi, Zengjie Zhang, Wanxin Jin, Marion Leibold, Martin Buss	2023	ITSC 2023	Accepted	No
KTH	Corridor MPC for Multi-Agents Inspection of Orbiting Structures	Gregorio Marchesini, Pedro Roque, Dimos Dimarogonas	2023	CDC 2023	Accepted	No
	Controller synthesis of signal temporal logic tasks for multi-agent systems via assume-guarantee contracts	Siyuan Liu, Adnane Saoud, Dimos Dimarogonas	2023	arXiv preprint: 2309.13499	Under submission	No
	Construction of Control Barrier Functions Using Predictions with Finite Horizon	Adrian Wiltz, Xiao Tan, Dimos V. Dimarogonas	2023	CDC2023	Accepted	No
	Communication-Guided Decomposition of Signal Temporal Logic Tasks through convex optimization	Gregorio Marchesini, Siyuan Liu, Dimos Dimarogonas	2023	Control engineering community	Under submission	No
	Distributed MPC for tractable Integer Programming of Multi-Agent Systems under separable Signal Temporal Logic tasks	Eleftherios E. Vlahakis and Dimos V. Dimarogonas	2023	Control engineering community	Under preparation	No

UU	Trustworthy driver-autonomous vehicle interaction: a participatory design study on the ethical considerations	Ana Tanevska, Ginevra Castellano and Katie Winkle	2023	Venue to be decided	Under preparation/ submission	No
	Communicating Awareness: Participatory Design Methods in Cognitive Human-Agent Interaction	Ana Tanevska, Arabinda Ghosh, Katie Winkle, Ginevra Castellano and Sadegh Soudjani	2023	HAI 2023 workshops	Under preparation/ submission	Yes
NLR						
SISW						

A.2 SymAware community building events

Partner	Event	Activity	Date & Location	Attendee
Report M01-M06				
MPI-SWS	Cyber-physical Systems Workshop 2023	Dr Sadegh Soudjani has given a talk on "Learning Approaches for Formal Synthesis of Cyber-Physical Systems" in line with the SwmAware project activities	March 21st 2023 at MPI-SWS, Kaiserslautern, Germany	Researchers and scientific members
	One-day workshop	Dr Sadegh Soudjani hosted Prof Raphael Jungers	2nd march 2023 at Newcastle University, UK	Researchers and scientific members
	Research collaboration	Dr Sadegh Soudjani visited the robotics Lab of Sheffiled University	January 4th and 5th 2023 at Sheffield, UK	Researchers and scientific members
	Presentation on learning methods for formal verification of CPS	Dr Sadegh Soudjani gave a presentation on learning methods for formal verification of CPS at Sheffield University	December 8th, 2022 at Sheffield, UK	Researchers and scientific members
	Presentation on learning methods for CPS	Dr Sadegh Soudjani gave an internal presentation on learning methods for CPS at MPI-SWS	November 8th, 2022 at MPI-SWS, Kaiserslautern, Germany	Researchers and scientific members
TU/e	Presentation on Data-Driven Verification for the Future of Mobility	Dr.ir. Sofie Haesaert gives a presentation at the Women in Data Science event in Eindhoven	March 27, 2023, High-tech Campus Eindhoven	Researchers, Engineers
	Presentation on Risk-Aware Task	Dr. Zengjie Zhang gives a presentation in the 42th Benelux meeting	March 23, 2023, Elspeet, The Netherlands	Researchers and Engineers

	Assignment with Formal Specifications for Heterogeneous Multi-Agent Systems			
	Research collaboration and presentation	Dr. Zengjie Zhang visited Technical University of Munich and gave a presentation on Safe Coordinate Control of Multi-Agent Systems.	November 11-15, 2022, Munich, Germany	Researchers and students
	Visit to Newcastle University in UK	Birgit van Huijgevoort visited Dr. Sadegh Soudjani and his student Oliver Schön	September 29 to October 5, 2022, Newcastle, UK	Researchers
KTH	Presentation on formal methods based control synthesis of CPS	Prof. Dr. Dimos Dimarogonas and Dr. Siyuan Liu attended and gave presentation at the 61st IEEE Conference on Decision and Control	December 6-9, 2022, in Cancun, Mexico	Researchers and scientific members
	Research collaboration	Dr. Siyuan Liu visited the Control Systems Group of Eindhoven University of Technology	December 18-22, 2022, in Eindhoven, Netherlands	Collaborators, researchers and scientific members
	Keynote speaker	Prof. Dr. Dimos Dimarogonas gave a talk on Multi-robot task planning and control at Sapienza University of Rome	December 22, 2022, in Rome, Italy	Researchers and scientific members
UU	Organization of the 2023 ACM/IEEE International Conference on	Prof. Dr. Ginevra Castellano served as the general chair of HRI 2023, with Dr. Katie Winkle as local chair and Dr. Ana	March 13-16, 2023, in Stockholm, Sweden	Researchers and scientific members

	Human-Robot Interaction (HRI 2023)	Tanevska as online and hybrid chair.		
NLR				
SISW				
Report M07-M12				
MPI-SWS	IFAC 2023	Dr Sadegh Soudjani organised a Workshop on the topic: Data-Driven Verification and Control in Cyber-physical systems	July 9-14, 2023 in Japan	Researchers and scientific members
	IFAC 2023	Dr Sadegh Soudjani gave a presentation on reinforcement learning for verification and synthesis of cyber-physical systems	July 9-14, 2023 in Japan	Researchers and scientific members
	Presentation for general professional stuff	Dr Sadegh Soudjani gave a presentation on the challenges of designing safety-critical systems.	May 30 2023 at Newcastle University	Researchers and scientific members
	Visit to Tokyo University	Dr Sadegh Soudjani visited Tokyo University in Japan and gave a presentation on reinforcement learning for verification and synthesis of cyber-physical systems	July 9-14, 2023 in Japan	Researchers and scientific members
	Yearly consortium meeting	The members of SymAware project attended the yearly consortium meeting	June 21-22, 2023 in in Eindhoven, Netherlands	Researchers and scientific members of SymAware project
TU/e	Presentation on Identifying	Dr. Zengjie Zhang gives a presentation to IEEE 26th	September 24-28, 2023, in Bilbao, Spain	Researchers and Industrial

	Reaction-Aware Driving Styles of SMPC Vehicles by Inverse Reinforcement Learning	International Conference on Intelligent Transportation Systems.		Engineers
	Presentation on Data-Driven Verification and Control Moving From Linear Dynamical Systems to Nonlinear Stochastic Systems	Dr. Sofie Haesaert gives a presentation at the 22nd World Congress of the International Federation of Automatic Control	July 9-14, 2023, at Yokohama, Japan	Researchers and Engineers
	Presentation on Safety Verification for AI-actuated Robotics	Dr. Zengjie Zhang gives an online presentation to University of Victoria	July 14, 2023, Online	Researchers and students
	Presentation on Optimal Experiment Design for Data-Driven Control and Verification	Dr. Sofie Haesaert gives a presentation at European Control Conference 2023	June 14-16, 2023, at Bucharest, Romania	Researchers and engineers
	Presentation on SySCoRe: Synthesis via Stochastic Coupling Relations	Birgit van Huijgevoort gives a presentation to 26th ACM International Conference on Hybrid Systems: Computation and Control	May 9-12 2023, at San Antonio, TX, USA	Researchers and Engineers
	Hosting visiting PhD	TU/e hosted a TUM PhD student Mrs. Ni Dang for	April 1 – 30, 2023, at Eindhoven, Netherlands	Researchers

	student from Technical University of Munich	one-month research visit.		
KTH	Presentation on Decentralized planning and control of multi-agent systems	Prof. Dr. Dimos Dimarogonas's group attended and gave presentation at the 22nd IFAC World Congress	July 9-14, 2023, in Yokohama, Japan	Researchers and scientific members
	Presentation on Human-in-the-loop planning and control of multi-robot systems	Prof. Dr. Dimos Dimarogonas's group attended and gave presentation at the 9th International Conference on Control, Decision and Information Technologies	July 03-06, 2023, in Rome, Italy	Researchers and scientific members
	Presentation on Multi-robot motion planning	Prof. Dr. Dimos Dimarogonas's group attended and gave presentation at the 2023 European Control Conference	June 14-16, 2023, in Bucharest, Romania	Researchers and scientific members
	Presentation on Distributed human-in-the-loop control of multi-robot systems	Prof. Dr. Dimos Dimarogonas's group attended and gave presentation at the 2023 IEEE International Conference on Robotics and Automation London	May 29-June 02, 2023, in London, UK	Researchers and scientific members
	Presentation on Cooperative sampling-based motion planning	Prof. Dr. Dimos Dimarogonas's group attended and gave presentation at the 2023 American Control Conference	May 31-June 02, 2023, in San Diego, CA, USA	Researchers and scientific members
	Multi-robot task planning	Prof. Dr. Dimos Dimarogonas gave a	April 21, 2023, Madrid, Spain	Researchers and scientific members

	and control under spatiotemporal specifications	keynote talk in Spanish Control Meeting		
UU				
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